



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SRI MURUGAN AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

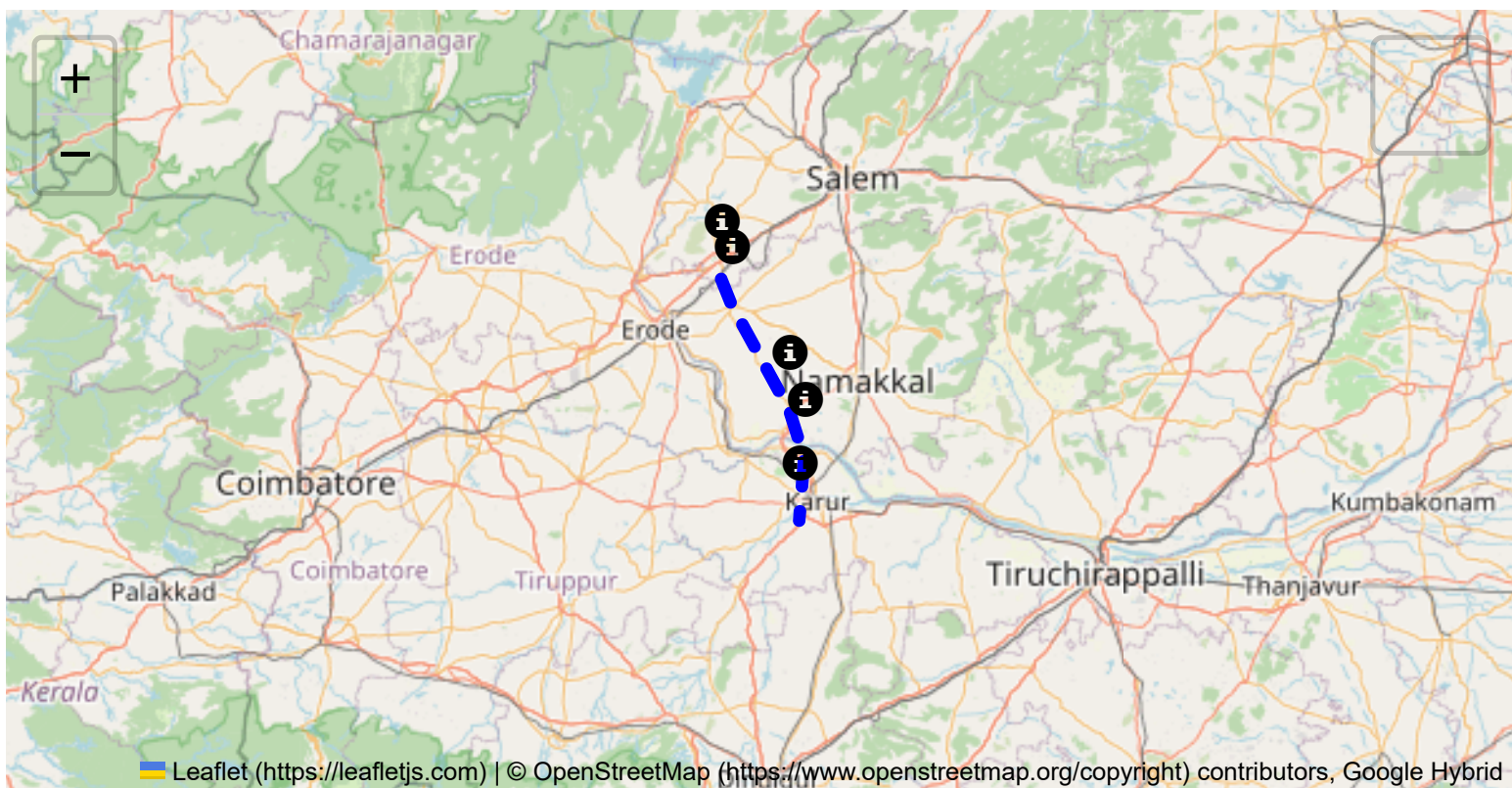
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

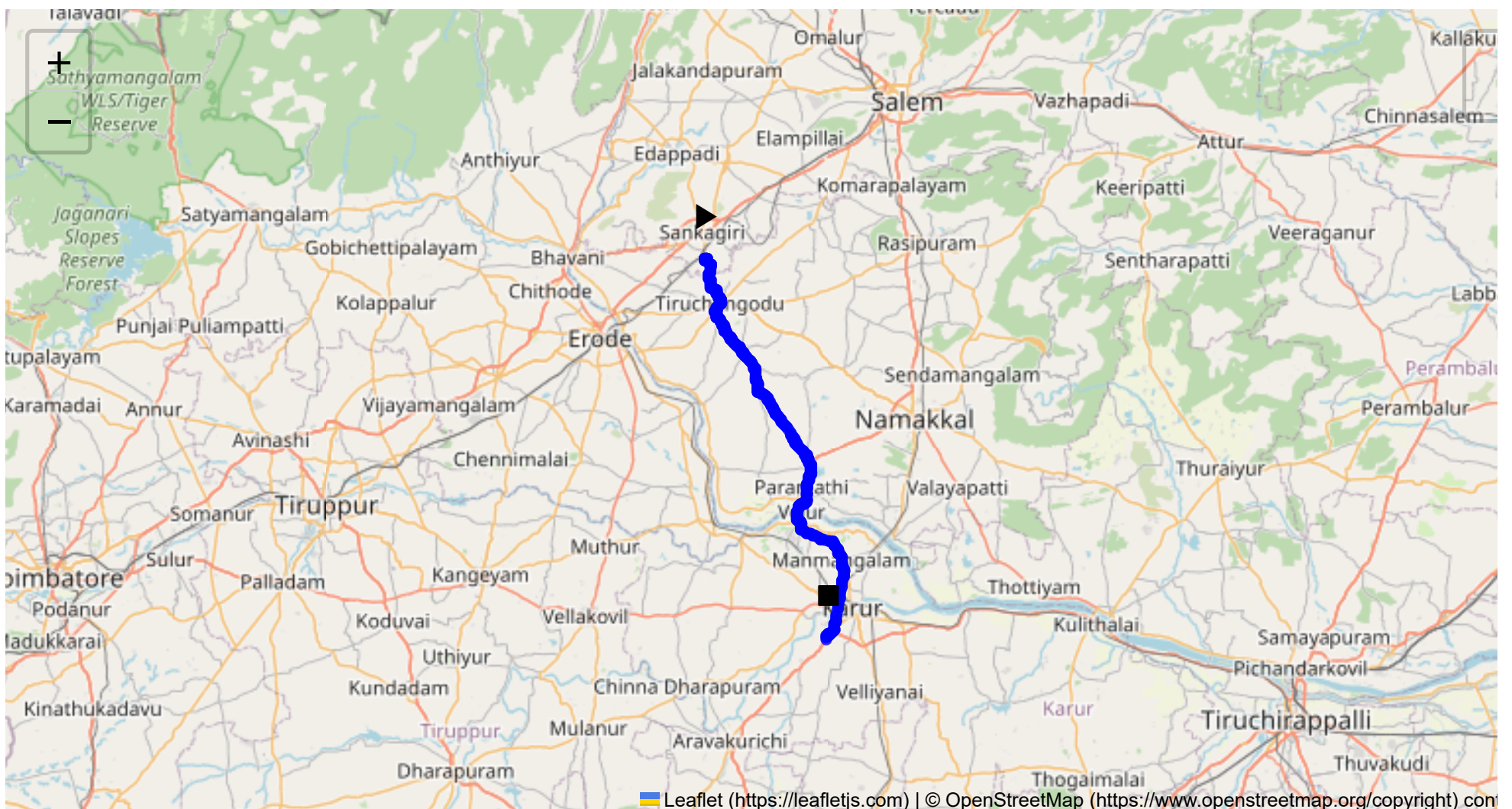
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 69.95 km
Estimated Duration: 1.4 hours
Adjusted Duration (Heavy Vehicle): 1.7 hours
Start: (11.4381, 77.8734)
End: (10.918041, 78.04349)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route travels from Sangagiri, Tamil Nadu to Pallapalayam, Tamil Nadu, covering approximately 69.95 kilometers. It passes through several urban and rural areas, primarily on state highways known for connecting smaller towns and regions. The major waypoints include Tiruchengode Central Bus Stand, SH 86 at Paramathi, and Karur.

2. Typical Weather Conditions and Potential Weather-Related Hazards

The region generally experiences a tropical climate with hot summers, moderate monsoons, and mild winters. During monsoon season (June to September), heavy rainfall might cause waterlogging and reduced visibility. Fog can be an issue in the winter months, potentially affecting early morning travel.

3. Analysis of Traffic Patterns

Traffic may be dense in the urban areas of Tiruchengode and Karur, with peak congestion typically occurring in the mornings between 8-10 AM and evenings from 5-8 PM. The roads connecting smaller villages may have lighter traffic but can still be busy during local market days or festive events.

4. Assessment of Road Quality and Infrastructure

The road quality along the state highways is generally good, but there may be potholes or uneven surfaces in certain sections, particularly in rural stretches. Expect narrow roads in village areas and standard traffic infrastructure like signals and pedestrian crossings in towns.

5. Suggestions for Alternative Routes for Emergencies

In case of a blockage on the primary route, consider using local roads connecting Paramathi to nearby towns such as Namakkal or Erode to re-route back towards Karur. However, these alternatives may involve minor roads which could be narrower or less maintained.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Transporting hazardous materials requires compliance with regional guidelines, which include restrictions on transporting through densely populated areas during peak hours. Drivers should carry appropriate documentation and must ensure adherence to loading and unloading regulations, especially within urban areas.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

There have been occasional incidents involving heavy vehicles on this route, predominantly due to poor road conditions during the monsoon or traffic congestion. However, specific cases involving hazardous materials are rare, pointing towards adequate enforcement of safety regulations.

8. Environmental Considerations and Sensitive Areas

The route passes near agricultural fields and small water bodies, which necessitates careful handling of hazardous materials to avoid environmental contamination. Crossing through towns requires heightened awareness to prevent accidents in populated areas.

9. Analysis of Communication Coverage

Mobile communication coverage in urban areas like Tiruchengode and Karur is strong. However, certain rural sections, particularly those not directly on major highways, may experience weak signals or dead zones.

10. Estimated Emergency Response Times

Urban areas offer quicker emergency response times, typically within 30 minutes. However, in more remote or rural sections of the route, response times may extend up to 60-90 minutes due to distance and road conditions.

11. Overall Summary of Risk Assessment

The route from Sangagiri to Pallapalayam via significant towns is moderately risky for transporting hazardous materials, mainly due to potential weather-related disruptions, traffic in urban centers, and areas with less reliable communication. Following local transport guidelines and monitoring weather forecasts are crucial for ensuring a safe journey. Adequate preparation and awareness can mitigate most risks associated with this route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Roundabout	High	11.38386, 77.89482	15 KM/Hr
2	U-Turn	High	11.3839218, 77.8950042	10 KM/Hr
3	Turn	Medium	11.43822, 77.87348	30 KM/Hr
4	Turn	High	11.43968, 77.87345	15 KM/Hr
5	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
6	Turn	High	11.37868, 77.89498	15 KM/Hr
7	Turn	High	11.37850, 77.89453	15 KM/Hr
8	Turn	Medium	11.16755, 78.01568	30 KM/Hr
9	Turn	High	11.15573, 78.02176	15 KM/Hr
10	Turn	High	11.14545, 78.02208	15 KM/Hr
11	Turn	High	11.14480, 78.02291	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
7	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium
16	hospital	Goverment Hospital	11.156647, 78.0213586	30 km/h	Medium
18	hospital	Government Hospital, Manmangalam	11.0275504, 78.0646437	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium
15	school	Sivabakkiam Muthusamy Matriculation Higher Secondary School	11.1788177, 78.0061696	30 km/h	Medium
17	school	Goverment Higher Secondary School	11.1511724, 78.0200723	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr
Coordinates: 11.38386, 77.89482



Risk Type: U-Turn
Risk Level: High
Speed Limit: 10 KM/Hr
Coordinates: 11.3839218, 77.8950042



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37868, 77.89498



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37850, 77.89453



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.16755, 78.01568



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.15573, 78.02176



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14545, 78.02208



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.14480, 78.02291
